

Symmetry as a Design Axis in Variational Quantum Learning

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Abstract—Variational quantum learning models are a promising application of near-term quantum computers, but their performance depends strongly on choosing an ansatz with a suitable inductive bias. Symmetry provides one such bias by enforcing that inputs related by task symmetries lead to consistent predictions. However, imposing symmetry does not fully determine the model architecture. In this work, we study the design choices that remain once a symmetry has been built into a variational quantum circuit. Using Tic-Tac-Toe board classification as a controlled benchmark with known symmetries and exact labels, we compare full board-symmetry sharing with relaxed symmetry constraints and then enrich the symmetric ansatz with trainable interactions aligned with the structure of the task. We find that relaxed symmetry preserves most of the generalization benefit, while the largest improvement comes from adding task-aligned interactions within the symmetric circuit. These results show that symmetry is not the end of ansatz design, but a starting point for choosing useful equivariant interactions.

Index Terms—quantum machine learning, geometric quantum machine learning, equivariance, symmetry, inductive bias, variational quantum circuits, parameter sharing, generalization

I. INTRODUCTION

Symmetry is one of the most useful things we can tell a model in advance. It marks the changes to the input that should not change the answer, so the model does not have to learn this for itself. Turn a picture of a cat sideways and it is still a cat; a model that already knows this need not waste data discovering it. The same idea underlies convolutional networks, which reuse one filter across an image [1], and geometric deep learning, which builds whole architectures around the transformations a task is supposed to ignore [2]. Variational quantum learning models (VQLMs) stand to gain the most from this discipline. They are parameterized quantum circuits trained as statistical models [3], [4], made expressive through data reuploading [5], [6] yet easily overwhelmed when that expressiveness is unconstrained. Embedding a known symmetry into the circuit resolves the tension: an equivariant ansatz routes input transformations into structured circuit transformations, making invariance a property of the architecture rather than a target of optimization, and improving generalization as a result [7].

We start where that result ends. Once symmetry is in the circuit, the usual remaining question is how much of it to enforce. We argue there is a second, more decisive question that is rarely asked: which symmetry-respecting interactions the circuit is given to learn. Treating symmetry as a design

space rather than a switch, we study both axes on a deliberately transparent benchmark, Tic-Tac-Toe board classification, where the relevant symmetry, the labels, and the patterns that decide the outcome are all known exactly and a fixed rule already attains perfect accuracy. The benchmark isolates the mechanism; the game itself is not hard.

We work through three layers. We first confirm that enforcing the full board symmetry improves generalization over an unconstrained circuit. We then relax that symmetry toward smaller subgroups and ask not just how much of the benefit survives, but whether a weaker symmetry can stay on par with the full one. Finally, and most importantly, we keep the symmetry but enrich it, adding trainable gates that act on the three-cell lines that actually decide the game, while still sharing their parameters across the symmetry. The takeaway generalizes well beyond the game: a known symmetry is not just a way to use fewer parameters. It is a language for choosing which interactions a quantum model is allowed to learn.

II. PROTOCOL

Each legal, reachable board is a list of nine values $g_i \in \{-1, 0, +1\}$, one per field, for circle, empty, and cross. Every board falls into one of three classes, circle win, draw, or cross win, written as target vectors $(+1, -1, -1)$, $(-1, +1, -1)$, and $(-1, -1, +1)$. Following the Meyer benchmark, unfinished games count as draws, and train and test splits are class-balanced so accuracy cannot be inflated by a dominant label. We enumerate all 5478 legal boards rather than sampling them, with the full procedure summarized in Appendix A. The label is invariant under the eight symmetries of the square, the four rotations and their reflections, which form the dihedral group D_4 . Figure 1(a) shows this directly: rotating or flipping a board never changes who has won.

To feed a board into the circuit, as in Figure 1(b), we place one qubit on each field and turn each value into a rotation angle $x_i = 2\pi g_i/3$ applied with a single-qubit R_X gate, so circle, empty, and cross land at three evenly spaced angles:

$$U(g) = \bigotimes_{i=0}^8 R_X \left(\frac{2\pi}{3} g_i \right).$$

Numbering the fields cyclically around the rim with the center last, as drawn in the panel, makes the board symmetry explicit. The four corners $\{0, 2, 4, 6\}$, the four sides $\{1, 3, 5, 7\}$, and the center $\{8\}$ are the three D_4 orbits: the symmetries shuffle positions within each set, but never mix one set with another.

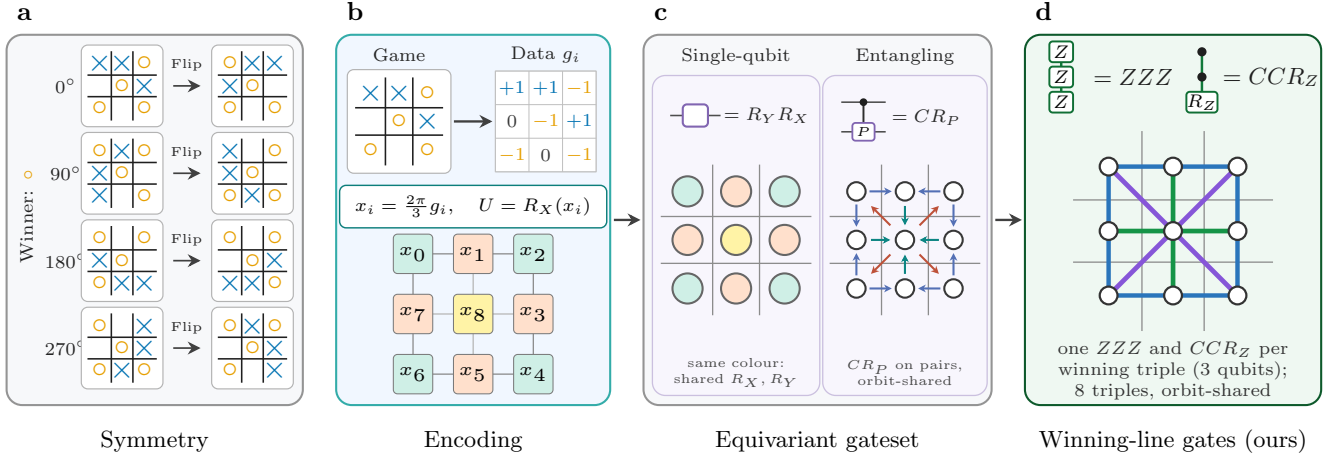


Fig. 1. From task symmetry to a task-aligned equivariant circuit. (a) A Tic-Tac-Toe label is invariant under the eight rotations and reflections of the square, the dihedral group D_4 ; the winner is unchanged throughout. (b) Each field carries a value $g_i \in \{-1, 0, +1\}$ for circle, empty, and cross, mapped to an angle $x_i = \frac{2\pi}{3}g_i$ and encoded on its own qubit via $R_X(x_i)$. Numbering the fields cyclically with the center last makes the D_4 orbits explicit: corners $\{0, 2, 4, 6\}$, sides $\{1, 3, 5, 7\}$, and center $\{8\}$. (c) The original edge ansatz: single-qubit $R_Y R_X$ rotations and controlled $C R_P$ entanglers on neighbouring fields and center connections, with gates in the same orbit sharing one parameter, shown by colour, so the model stays equivariant. (d) Our extension adds orbit-shared gates on the eight winning triples, rows, columns, and diagonals: one ZZZ rotation and one symmetric CCR_Z per triple, enriching the circuit with the motifs that decide the game without breaking the symmetry. Panel layout adapted from Meyer et al. [7].

The circuit is a stack of identical layers. Each layer reencodes the board with $U(g)$, a data re-uploading step that lets the circuit express nonlinear functions [5], and then applies a trainable block. Unless stated otherwise, we use $L = 3$ layers with $p = 2$ repetitions of the block per layer. The block is the original edge ansatz shown in Figure 1(c): a single-qubit $R_Y R_X$ rotation on every qubit, followed by controlled $C R_P$ entanglers on neighbouring fields and center connections.

Symmetry enters through how these gates are parameterized. Instead of giving each gate its own angle, all gates within an orbit share one trainable parameter, indicated by colour in Figure 1(c): corners share one single-qubit rotation, sides another, the center keeps its own, and the $C R_P$ entanglers are tied across the orbits of the directed pairs they act on. A rotated or reflected board is then never a new, unrelated input; it permutes the same qubits and reuses the same parameters, which is precisely what makes the model equivariant. The readout follows the same geometry: we average the Pauli- Z expectation over each orbit to obtain a three-number vector for corners, center, and sides, and take its largest component as the predicted class.

Because the orbits are defined by whichever symmetries we choose to enforce, we can dial the symmetry up or down without touching the wiring. For a subgroup $H \leq D_4$, parameters are tied across the orbits induced by H , leaving the circuit topology fixed and changing only the sharing pattern. Comparing no sharing, the two Z_2 subgroups, C_4 , D_2 , and full D_4 gives 204, 108, 126, 60, 78, and 54 trainable parameters for the edge ansatz at $L = 3, p = 2$, respectively; in this

family, more symmetry means fewer free parameters.

The edge ansatz only couples neighbouring fields, yet a win is decided by a full three-cell line. This gap motivates the extension in Figure 1(d). We add trainable gates directly on the eight winning triples, the three rows, three columns, and two diagonals shown in blue, green, and purple, using one ZZZ rotation and one symmetric CCR_Z interaction per triple. We test each channel on its own and together. Our main model, edge+lines, keeps the original edge ansatz and adds both line-gate channels. Crucially, these new gates are shared over the same orbits, so the model remains equivariant under the chosen subgroup. At $L = 3, p = 2$, the full- D_4 edge model has 54 parameters and the full- D_4 edge+lines model has 90, still far below the 204 of the unshared edge baseline.

All reported runs use mean-squared error, Adam with learning rate 0.01 [8], batch size 15, 30 minibatch updates per epoch, 100 epochs, fixed seed blocks, PennyLane simulation [9], and PyTorch optimization [10]. Exploratory runs outside this protocol are not used for any headline claim.

III. RESULTS

Figure 2 shows the main empirical story. In panel (a), the unconstrained edge ansatz carries many more parameters yet generalizes worse than the D_4 -equivariant edge model. At 600 training examples, edge/none reaches about 0.63 test accuracy while edge/ D_4 reaches about 0.69 with a smaller train-test gap. This reproduces the central lesson of Meyer et al.: the correct board symmetry removes irrelevant functions from the hypothesis class rather than simply shrinking the model.

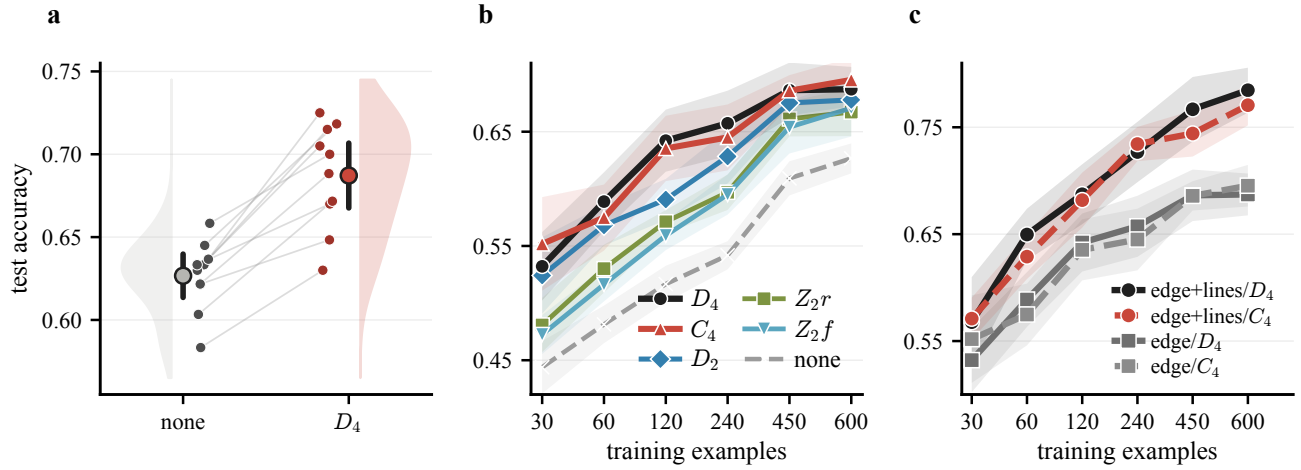


Fig. 2. Main results. (a) Full D_4 improves over no sharing. (b) C_4 closely follows D_4 across train sizes. (c) Edge+lines gives the largest gain while preserving equivariance. Error bars are 95% confidence intervals over seeds.

Panel (b) makes the picture less binary. Full D_4 is the conservative default, but C_4 tracks it closely across the data grid: it is slightly ahead at 30 examples, and from 120 through 600 examples the two stay essentially together while weaker or unconstrained sharing lags. We read this cautiously. It does not establish that partial symmetry is generally better; it shows that symmetry strength is a tunable bias, with rotation-only equivariance preserving most of the benefit while relaxing reflection constraints. The random-sharing control in fig. 3(a) is the guardrail: it fixes parameter count but scrambles the group-orbit structure, and performs substantially worse than orbit-based sharing at every subgroup. The useful prior is the board geometry, not just the smaller circuit.

The strongest effect, in fig. 2(c), comes from enriching the symmetry rather than weakening it. The original edge/ D_4 model sits near 0.69; adding both winning-line channels to the same equivariant circuit lifts the best D_4 model to roughly 0.80, with the C_4 variant close behind. This is the central result: the gain comes not from abandoning symmetry, but from giving the equivariant ansatz direct access to the motifs that decide the label.

Figure 3 rules out two simpler explanations. The gain is not arbitrary parameter reduction, since matched-size random sharing falls short of orbit sharing in panel (a). It is not simply more parameters either: edge+lines/ D_4 has 90 parameters, more than edge/ D_4 at 54 but far below the 204 of the unshared baseline. Panel (c) makes the point geometric: accuracy does not follow parameter count, and edge+lines models occupy a higher band at modest size. The ablation in panel (b) confirms the composition: line-only circuits help, but the strongest model combines local edge structure with both winning-line channels.

IV. DISCUSSION

Our results support an architectural view of equivariance in variational quantum learning. Meyer et al. showed that

incorporating task symmetry into a VQLM can improve generalization. We take this as a starting point and ask what remains to be designed once that symmetry has been imposed. The answer is that equivariance is not a complete ansatz specification: it constrains parameter sharing and responses to symmetry-related inputs, but not which trainable interactions the circuit should contain.

The subgroup experiments do not show that rotation-only equivariance is generally preferable to full board equivariance. They show that symmetry strength is a meaningful modeling choice. In this benchmark, weaker symmetry keeps most of the benefit of the fully symmetric model while giving the circuit slightly more freedom. Subgroup comparisons can therefore diagnose overconstraint when a fully equivariant circuit underperforms for a given data regime, circuit family, or optimizer.

The stronger empirical effect comes from changing the interactions inside the equivariant ansatz. The original edge circuit couples neighboring fields and must represent winning triples indirectly. The line interactions act directly on the three-cell structures that determine the label, while still sharing parameters across symmetry-related instances. This gives the largest improvement in our experiments. The matched random-sharing control rules out parameter reduction as the explanation: what matters is that trainable degrees of freedom are placed according to the task geometry.

This suggests a practical design principle. Once a task symmetry has been identified, one should ask not only whether the circuit respects it, but whether the equivariant gates expose the structures through which the label is determined. Equivariance is a starting point for ansatz design rather than its end point: it defines a constrained design space in which the choice of interactions can still be decisive.

Tic-Tac-Toe is small, enumerable, and exactly solvable by a deterministic rule oracle, so its value is as a controlled setting for isolating symmetry-aware architectural choices. The supported claim is narrow: task-aligned equivariant interac-

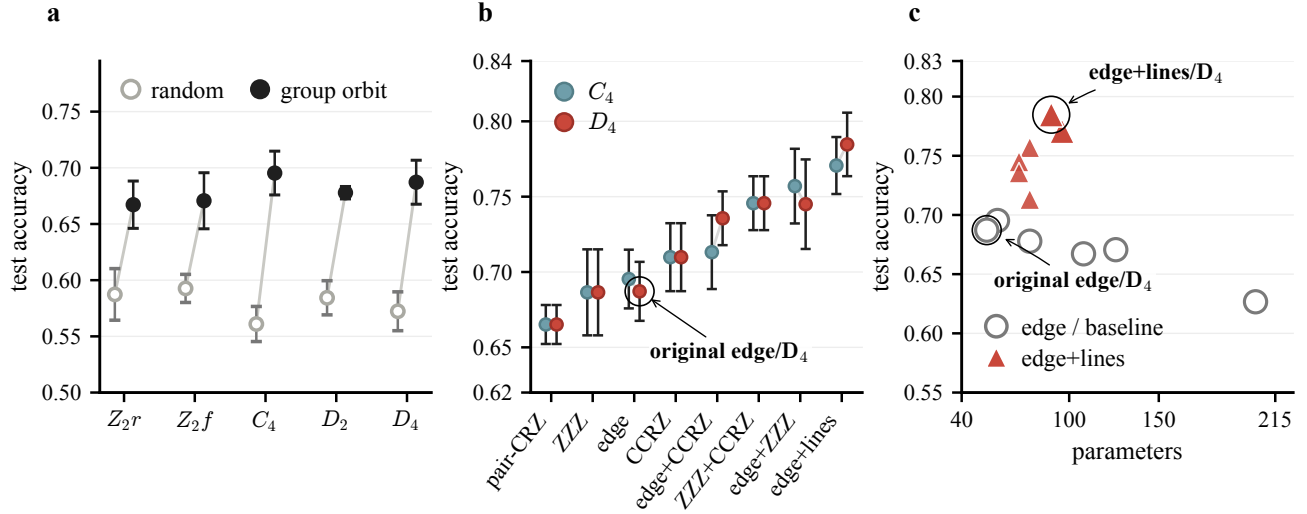


Fig. 3. Controls and ablations. (a) Parameter-matched random sharing underperforms group-orbit sharing. (b) Ablations are anchored by the labelled original edge/ D_4 ansatz; the strongest model combines edge interactions with both line channels. (c) Accuracy is not explained by parameter count alone.

tions can improve generalization while preserving the imposed symmetry. For larger problems, the transferable recipe is to identify task-relevant motifs and encode them as trainable interactions shared over the corresponding symmetry orbits.

A natural next step is to compare enriched equivariant ansatzes with soft symmetry approaches, where controlled symmetry-breaking terms interpolate between hard equivariance and a free ansatz. Such comparisons should include not only the original symmetric circuit, but also equivariant circuits whose interactions have been designed to match the task. The broader lesson is that symmetry-aware quantum circuit design should use symmetry to organize the trainable interactions through which the model learns.

APPENDIX A: DATA GENERATION

Data generation and sampling. We follow the Tic-Tac-Toe benchmark protocol of Meyer et al. Legal reachable board states are generated by expanding the game tree from the empty grid, alternating cross and circle moves, and stopping once a game is decided. This yields 5478 distinct boards, stored as cyclically indexed values $g_i \in \{-1, 0, +1\}$ with the center last so the corner, side, and center fields match the D_4 orbits. Labels follow the winning-line rule, with unfinished positions counted as draws. For every seed and training size we draw class-balanced train and test subsets and reuse the same split for all models compared at that setting. Meyer et al.’s headline Tic-Tac-Toe setup uses 450 training examples and a fixed test set of 600 examples; our sweep keeps the same balanced-sampling rule and test size while varying the number of training examples.

Reproducibility. Code, figure sources, and checked CSVs are available at github.com/eybmits/qc_symmetry. All reported VQLM runs use one protocol: $L = 3$, $p = 2$, the paper single-qubit block, mean-squared error, Adam

at learning rate 0.01, batch size 15, test size 600, train sizes 30, 60, 120, 240, 450, 600, seeds 0–9, and 100×30 optimizer updates. Figures regenerate with `python paper/make_figures.py`; the manuscript rebuilds with `latexmk -pdf -interaction=nonstopmode main.tex`. Validation requires 360 edge rows, 120 edge+lines rows, 160 ablation rows, and 110 random-sharing rows; intervals are 95% confidence intervals, and the rule oracle is a label sanity check, not a plotted baseline.

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